Mahmoud Bekri PhD, CIB

Research Economist, Islamic Development Bank Institute, Jeddah

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PROFILE

Research Economist at the Islamic Development Bank Institute, trained in physics and in finance at KIT, with more than ten years in the IsDB Group. His research covers heavy-tailed distributions, coherent risk measures (CVaR and AVaR under α-stable laws) and Islamic finance, and is applied in risk modelling, Sukuk and product design, and capacity building for central banks and regulators. In 2026 he is seconded to Model Risk in the IsDB Risk Management Department.

EDUCATION

2014 — **PhD in Economics and Finance**
Karlsruhe Institute of Technology (KIT), Germany · Institute of Economics (ECON), Chair of Econometrics and Statistics
"Portfolio Optimization and Risk Management under Non-normal Stable and Tempered Stable Distributions in Islamic Finance"
Magna cum laude · Supervisors: Prof. Svetlozar Rachev (Stony Brook University, US), Prof. Wolf-Dieter Heller (KIT), Prof. Frank Fabozzi (Yale University, US)

2011 — **German Diplom in Physics (Diplom-Physiker)**
Karlsruhe Institute of Technology (KIT), Germany
"Studies of Si-MCM-41 and Al-MCM-41 Surface Structure and Surface Charge"
Final grade "very good" (sehr gut) · Supervisor: Prof. Schimmel

Professional certificates

2024 — **Certificate in Advanced Project Management**
Meirc Professional Certificate (MPC), member of PMI

2018 — **Leadership Certificate in Islamic Banking and Finance**
Cambridge Islamic Finance Leadership Programme (IFLP) 2018

2016 — **Certified Islamic Banker (CIB)**
CIBAFI

SELECTED PUBLICATIONS

Book and book chapter

Bekri, M. (2021). *Heavy Tails and Coherent Risk Measures in Islamic Finance.* Islamic Development Bank Institute, Jeddah. 170 pp. ISBN 978-603-8306-06-2.

Bekri, M., Kabir Hassan, M., Alam, N. (2017). On the dependency structure of Islamic assets. In M. Kabir Hassan (ed.), *Handbook of Empirical Research on Islam and Economic Life*, pp. 602–623.

Doctoral thesis

Bekri, M. (2014). *Portfolio Optimization and Risk Management under Non-normal Stable and Tempered Stable Distributions in Islamic Finance.* Doctoral thesis, KIT, Germany.

Indexed journals

Hajjar, M., Bekri, M. (2026). Quick Commerce and Islamic Ethics: Maqāṣid and Policy Assessment. *International Journal of Islamic Finance and Sustainable Development*, 18(3), 41–59. doi:10.55188/ijifsd.v18i3.1461

Mirzaei, M., Bekri, M. (2017). Energy consumption and CO_2 emissions in Iran, 2025. *Environmental Research*, 154, 345–351.

Mirzaei, M., Bekri, M., Ramakrishnan, S. (2016). Corporate Default Prediction with Industry Effects: Evidence from emerging markets. *International Journal of Economics and Financial Issues*, 6(S3), 161–169.

Ramakrishnan, S., Mirzaei, M., Bekri, M. (2015). Corporate Default Prediction with Adaboost and Bagging Classifiers. *Jurnal Teknologi*, 73(2), 45–50.

Ramakrishnan, S., Mirzaei, M., Bekri, M. (2015). A Multi-Industry Default Prediction Model using Logistic Regression and Decision Tree. *Research Journal of Applied Sciences, Engineering and Technology*, 9(10), 856–861.

Ramakrishnan, S., Mirzaei, M., Bekri, M. (2015). Adaboost Ensemble Classifiers for Corporate Default Prediction. *Research Journal of Applied Sciences, Engineering and Technology*, 9(3), 224–230.

Bekri, M., Kim, Y. S. (2015). Tail risk analysis of the S&P/OIC COMCEC 50 index. *Borsa Istanbul Review*, 15(1), 1–16.

Bekri, M., Kim, Y. S. (2014). Portfolio management with heavy tailed distributions in Islamic Finance. *Journal of Islamic Economics, Banking and Finance*, 10(2), 114–137.

Bekri, M., Kim, Y. S., Rachev, S. T. (2014). Tempered stable models for Islamic finance asset Management. *International Journal of Islamic and Middle Eastern Finance and Management*, 7(1), 37–60.

Conference papers

Bekri, M. (2026). Pricing Catastrophe-Tailed Energy Assets: A Stable-Law Expected-Shortfall Correction to the Cost of Capital for Nuclear Power. Accepted, 4th IAEE MENA Conference, KAPSARC, Riyadh, 7–8 December 2026.

PROFESSIONAL EXPERIENCE

01/2026–present

Secondment, Model Risk

Risk Management Department, IsDB

- Contribute to building the Bank's model risk management framework.
- Designed a multi-manager equity risk oversight framework that monitors external equity managers against their benchmark (tracking error, risk attribution and performance assessment), documented for formal model validation.
- Built an asset-liability management cash-flow stress model and an Islamic ALM model, with full reconciliation and quality-control checks.
- Supported hedging analysis, including swap hedge accounting, and delivered an internal workshop on hedging Islamic equity portfolios.

2018–present

Research Economist

Islamic Development Bank Institute (IsDBI)

- Lead research, policy advisory and capacity-building initiatives to advance Islamic finance across IsDB member countries.
- Conduct applied economic and quantitative research on sustainable finance, risk-sharing instruments and fintech innovation, and develop financial models for Islamic products and sovereign instruments such as Sukuk.
- Collaborate with central banks, ministries of finance and development partners, and represent the Institute at international forums and conferences.

Notable achievements

- **Book:** authored *Heavy Tails and Coherent Risk Measures in Islamic Finance* (IsDBI, 2021).
- **Patents and products:** played a key role in IsDBI's "Smart Voucher System" (US Patent 12282916) and contributed to the patents "The Blockchain-based Voucher System" and "Blockchain-based Credit Management"; worked on the Smart Economy Project and designed a fintech product for sovereign Islamic finance initiatives; pricing analyst and product developer for the Sukuk Assets Utility Awqaf.
- **Benchmark:** led a task force and performed the quantitative work on "Developing an Islamic Financial Returns Benchmark for Islamic Financial Institutions", including the database and analysis of the Consensus Economics survey data.

- **Capacity building:** designed and delivered programmes for central banks and regulators in Libya, Morocco, Uganda and Mauritania; represented IsDBI in the Libya Task Force (Central Bank of Libya, Libya Development Bank); delivered "Introduction to Islamic Finance" for the PTCI programme in Dakar, Senegal; conducted an international workshop on capital regulation for Islamic financial institutions in Oman; member of the Center of Excellence on Islamic Finance team that organised its first workshop in Madinah.
- **Digital currency and crypto assets:** coordinated the Palestine Monetary Authority Digital Currency Project and was on the organising team of the first Economists' Forum, "The Rise of Central Bank Digital Currency: Motives, Approaches, and Techniques"; presenter and panellist at the ADFIMI and Qatar Development Bank workshop on "Sharia Compliance Issues of Crypto Assets".
- **Grants, projects and country work:** contributed to several Islamic finance grants (Awqaf, participatory and social finance); drafted the IsDBI contribution to the Member Country Partnership Strategies for Niger, Guinea and Morocco; technical assistance project officer for Uganda, Burkina Faso and the UAE; evaluator of the Queen Alia Airport Project, Amman, with the IDB Operations Evaluation Department; co-editor of the Islamic Finance Country Report on Djibouti.
- **Research papers:** "Taming Inflation: An Islamic-finance Perspective"; "The Determination of Alpha in the Capital Adequacy Ratio for Islamic Financial Institutions" (TICIEF 2019); "On the Role of Cryptocurrencies in Islamic Finance, Real Economy and Sustainability" (Istanbul); research on GDP-linked Sukuk and post-COVID-19 inflation; a case study of the Islamic Banking Africa Summit.
- **Blogs and translation:** authored blogs including "Inflation Dependence Structure in a post-COVID-19 World" and "Crypto GDP Sukuk, for Better Debt Servicing in Post-COVID-19", and translated IsDBI works between English, Arabic and French.

2015–2018

Young Professional

Islamic Development Bank Group, Young Professionals Programme (YPP): training and three rotations

- **Financial Product Development Center** (second and third rotations): represented the department in the IDB Group Risk Management Committee; authored studies on GDP-linked and commodity-linked structures, credit default swaps and their modelling, and fiduciary and third-party assessment risk; took part in designing the Sovereign Finance Company and its cash-flow model; optimised the stock basket of exchangeable Sukuk; built a Sukuk databank and analysed its tail risk; analysed the bonds used to price Sukuk; drafted a first export index for OIC member countries.
- **Islamic Financial Services Department** (first rotation): reviewed the feasibility study of the Awqaf Properties Investment Fund Project in Cotonou, Benin, and joined its appraisal mission; presented the Zitouna Tamkeen Project to the investment committees; audited the Saudi Project for Hajj Meat (Adahi); contributed to the technical assistance process review.
- Attended the 29th International Summer School of the Swiss Association of Actuaries on Quantitative Risk Management, Lausanne.

11/2013–02/2015

Postdoctoral Researcher

Institute for Economic Theory and Statistics (ETS), KIT, Germany

- Research on financial engineering, portfolio optimisation and risk management; managed the research projects on Islamic finance.
- Taught financial engineering and advanced financial methods; supervised bachelor's and master's theses and co-supervised a PhD thesis.

11/2013–09/2014

Senior Consultant, Islamic finance projects

FinAnalytica and Bravo Risk Management, New York

- Built portfolio analytics and a risk management tool for daily trading; applied stable and heavy-tailed distributions to DAX components and Sukuk to improve portfolio performance and control tail risk.

Earlier roles

04/2013–10/2013	**Senior Social Network Analysis Analyst and Consultant**, SONEAN GmbH, Frankfurt: SNA-based business strategies and analysis of decision-makers' networks.
12/2012–02/2013	**Sales Manager**, Edmund Optics, Karlsruhe: technical advice, feasibility analysis and offers.
09/2012–11/2012	**Intern, Portfolio Pricing**, Hoist Finance Group, Düsseldorf: pricing of non-performing loan portfolios.
03/2011–08/2012	**Project Manager**, TISFO (Tunisian Industrial Services for Oil), Sousse: market research, project delivery and budget control.

REFEREE FOR

2015	*International Journal of Islamic and Middle Eastern Finance and Management*
2014	*Frontiers in Mathematical Finance*

LANGUAGES AND SOFTWARE

Languages	Arabic (mother tongue) · English, French and German (fluent)
Software	MATLAB, Python, MS Office
Interests	Reading, writing, sport and travelling

OTHER ACTIVITIES

- Member of the German Business Circle of the German-Saudi Arabian Liaison Office for Economic Affairs, Jeddah, Saudi Arabia.
- Member of the Tunisian Scientific Society (TWG), Karlsruhe, Germany.
- Member of the academic "Wednesday Talk", a meeting of economic scholars at King Abdulaziz University, Jeddah, Saudi Arabia.
- Former Chairman of the Jury of the Master of Islamic Financial Management, Effat College of Business, Jeddah, Saudi Arabia.